MILITARY SPECIFICATION

CABINET, ELECTRICAL EQUIPMENT CY-7887/UPQ-5

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide rack mounting and interconnecting cabling for selected units of the AN/UPQ-5 Mission Recorder Display Set and for items of Peculiar Ground Support Equipment associated with the AN/UPQ-5 system. Additionally, the equipment shall provide non-operating storage facilities for other items of PGSE associated with the AN/UPQ-5 and for selected spare Shop Replaceable Assemblies from the AN/UPQ-5.

1.2 Classification - The equipment covered by this specification shall consist of a single movable console containing the following replaceable assemblies:

Designation	Item	Applicably Paragraph
--	Cabinet Subassembly	3.5.1
A1	PGSE Storage & Power Panel Assembly	3.5.2
--	Disc Storage Shelf Assembly	3.5.3
A2	Mission Recorder Mount	3.5.4
--	SRA Storage Drawer Assembly	3.5.5
A3	Electronic Equipment Air Cooler	3.5.6
--	Blank Panels	3.5.7

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in paragraph 6.7.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue shown or the latest issue where no date is listed form a part of this specification to the extent specified herein. The contractor may use later issues of these documents at his option. However, if such use results in a change in design, performance, or interchangeability; approval of the procuring activity shall be obtained.

SPECIFICATIONS

Federal

W-C-596/13B 31 Oct 1977	Connector, Plug, Electrical, General Purpose, Hospital Grade Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz

Military

MIL-C-915E Amend. 1 Supp 1 5 April 1973	Cable And Cord Electrical, For Shipboard Use General Specification For
MIL-G-3767B Amend. 2 Supp 1A 27 April 1970	Connector, Plug And Receptacle (Power, Blade Type), General Specification For
MIL-E-16400G Amend. 1 1 December 1976	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specification For
MIL-E-17555G Amend 2 15 April 1970	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-T-18303B 1 September 1966	Test Procedures, Preproduction, Acceptance, And Life For Aircraft Electronic Equipment, Format For
MIL-N-18307F 4 August 1978	Nomenclature And Identification For Electronic, Aeronautical, And Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-28800B Amend. 1 20 July 1977	Test Equipment For Use With Electrical And Electronic Equipment, General Specification For

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MIL-C-39012B Amend. 2 Supp 1B 14 October 1976	Connector, Coaxial, Radio Frequency, General Specification For
MIL-C-39019B Amend. 2 Supp 1 19 July 1978	Circuit Breakers, Magnetic, Low-power, Sealed, Trip Free, General Specification For
MIL-A-55339 Amend. 4 Supp 1B	Adapter, Connector, Coaxial, Radio Frequency, (Between Series And Within Series), General Specification For
MIL-R-81687	Recorder Set, Mission AN/USH-17(V)
MIL-D-81688A	Display Set, Mission Recorder AN/UPQ-5B
MIL-D-81976 17 June 1974	Dummy Load, Electrical DA-662/UPQ-5
MIL-S-81977 17 June 1974	Simulator, Radar Signal SM-697/UPQ-5

Naval Air Systems Command

AR-10A 1 January 1969 Notice 1 16 June 1969	Maintainability Of Avionics Equipment And Systems General Requirement For

Military

MIL-STD-167-1 1 May 1974	Mechanical Vibration Of Shipboard Equipment (Type I-Environmental And Type II-Internally Excited)

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MIL-STD-785A 28 March 1969 Notice 2 15 June 1979	Reliability Program For Systems And Equipment Development And Production
MIL-STD-794D 15 December 1972 Notice 4 8 August 1975	Part And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810C 10 March 1975	Environmental Test Methods
MIL-STD-882A 28 June 1977	System Safety Program Requirements
MIL-STD-965 15 April 1977 Notice 1 22 December 1978	Parts Control Program
DOD-STD-1399/300	Interface Standard For Shipboard Systems Section 300, Electric Power, Alternating Current
MIL-STD-1472B 31 December 1974 Notice 2 10 May 1978	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MS-3102C 15 May 1969	Connector, Receptacle, Electric, Box Mounting
MS-3106B 15 May 1969	Connector, Plug, Electric, Straight
MS24266K 14 November 1977	Connector, Plug, Electrical Straight, Miniature

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DRAWINGS

HRB-Singer, Inc. (FSCM 99592)

325-185	Converter, Frequency, Static CV-2933A/UPQ-5, Source Control Drawing For
335-804	Mounting Kit, Rack, Specification Control Drawing For
601209	Mission Recorder Display Set AN/UPQ-5, Engineering Drawing Package For (Dwg 601209, subordinate drawings, and associated lists)
603680	Radar Signal Simulator SM-697/UPQ-5, Engineering Drawing Package For (Dwg 603680, subordinate drawings, and associated lists)
603685	Electrical Dummy Load DA-662/UPQ-5, Engineering Drawing Package For (Dwg 603685, subordinate drawings, and associated lists)
610600	Video Indicator IP-1314/UPQ-5, Engineering Drawing Package For (Dwg 610600, subordinate drawings, and associated lists)

RCA Corp (FSCM 49671)

8370986	Mission Recorder Set AN/USH-17(V), Engineering Drawing Package For (Dwg 8370986, subordinate drawings, and associated lists)

PUBLICATIONS

Naval Air Systems Command

NAVAIR 16-30UPQ5-1	Technical Manual, Operation And Maintenance Instructions (Intermediate) Display Set, Mission Recorder, AN/UPQ-5D Part Number 601209-5

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NAVAIR 16-30UPQ5-4	Technical Manual, Operation And Maintenance Instructions With Illustrated Parts Breakdown (Intermediate) Indicator, Video, IP-1314/UPQ-5 Part Number 610600-1
NAVAIR 16-30USH17-1	Technical Manual, Maintenance Instructions (Intermediate AN/USH-17(V)3 And AN/USH-17(V)7 Mission Recorder Sets
NAVAIR 16-30USH17-2	Technical Manual, Illustrated Parts Breakdown (Intermediate) AN/USH-17(V)3 And AN/USH-17(V)7 Mission Recorder Sets
NAVAIR 16-35SM697-1 (Formerly NAVAIR 16-45-1939)	Technical Manual, Operation And Maintenance Instructions With Illustrated Parts Breakdown (Intermediate) Simulator, Radar Signal SM-697/UPQ-5 Part Number 603680-1 and Dummy Load, Electrical DA-662/UPQ-5 Part Number 603685-1

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic Technology shall be considered, and microelectronic items shall conform to requirements specified herein. When used, microelectronic devices shall have two or more sources, and the device from each source shall be interchangeable. Interchangeability shall be demonstrated by random installation of devices from two or more sources during fabrication of production quantities. Successful operation of the equipment with a random mixing of device sources shall be proof of interchangeability. Unsuccessful operation of the equipment shall be evaluated to determine if a device is faulty or is not interchangable.

(2) Except as otherwise specified herein, all other parts and material requirements shall conform to MIL-E-16400.

(3) Nonrepairable subassemblies shall be used in accordance with paragraph 3.1.3(2) of Specification AR-10.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Material Approval - Approval for use of nonstandard parts and materials (including electron tubes, transistors, diodes and microelectronics devices) shall be obtained as outlined in specification MIL-E-16400. Nonstandard parts data shall be in accordance with applicable requirements of MIL-STD-965.

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3.2.2 Modules - Electronic portions of the equipment shall be functionally modularized using the requirements of AR-10 as a guide.

3.3 Design and Construction - The design, construction, and workmanship of the equipment shall be, wherever feasible, consistent with the design, construction, and workmanship requirements for the AN/UPQ-5 Mission Recorder Display Set as described in MIL-D-81688. Where MIL-D-81688 is not applicable, the design, construction, and workmanship requirements of MIL-E-16400 shall be used as a guide except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 650 pounds.

3.3.2 Reliability - When specified in the contract or order, the contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - When the contract or order requires a reliability program under provisions of section 3.3.2 herein, the mean (operating) time between failures (MTBF) in hours and the test requirements shall also be specified in the contract or order.

3.3.2.4 Time Totalizing Meter - Time totalizing meters are not required.

3.3.3 Cabling and Connectors - Cabling and connectors shall meet the following requirements:

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3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-16400 except as otherwise specified herein.

3.3.3.2 Interconnecting Cabling - The equipment shall be capable of satisfactory operation using external power cabling in accordance with the applicable requirements of MIL-C-915. The external power cabling shall be unshielded, except that a minimum number of individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. The external power cable and that portion of the connector attached to the cable shall be supplied as part of the equipment.

3.3.4 Control Panels - All rack and console mounted control panels shall conform to the applicable requirements of MIL-E-16400. The configuration of all control panels must be approved by the procuring activity prior to the Preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-16400.

3.3.6 Electromagnetic Interference Control - The requirements of MIL-E-16400 shall be used as a guide in the design of the equipment for the control of electromagnetic interference. As a minimum, the following requirements shall apply.

3.3.6.1 Enclosure - The equipment cabinet (enclosure) shall be in accordance with the requirements of section 3.5.1.2 herein.

3.3.6.2 EMI Gasketing - EMI gasketing shall be provided as specified in section 3.5.1.3 herein.

3.3.6.3 Power Line Filters - EMI filters shall be provided on the power input line to the equipment. These filters shall provide a minimum insertion loss, at their rated load, of 30 dB at 15 KHz, increasing to a minimum of 80 dB for frequencies between 60 KHz and 1 GHz.

3.3.7 Provisions for Maintainability - Construction and packaging shall be as outlined in paragraphs 3.1.2, 3.1.4 and 3.1.5 of Specification AR-10.

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3.3.7.1 Organizational Level Maintenance - The design of the equipment shall be such as to permit maintenance at the organizational level to determine that the equipment meets its performance requirements and is satisfactory for its intended use.

3.3.7.2 Weight of Replaceable Assemblies - The weights of individual assemblies shall not exceed the weights specified in section 3.5.

3.3.7.3 Accessibility - To provide equipment accessibility, unit mounting shall be in accordance with the following:

(1) Power Panel (A1) - The PGSE Storage and Power Panel Assembly shall be a standard 19-inch rack mounted assembly secured by front panel fasteners. Weight bearing supports shall be provided in the equipment rack as necessary.

(2) Recorder Mount (A2) - The Mission Recorder Mount shall be a slide-mounted standard 19-inch rack mounted unit secured by front panel fasteners.

(3) Blower (A3) - The Electronic Equipment Air Cooler shall be a standard 19-inch rack mounted unit secured by front panel fasteners. Additional rear-attaching mounting hardware is permissible.

(4) Power Strip (A4) - The Electrical Outlet Strip shall be mounted in the inside rear of the equipment cabinet to permit AC power cords from front mounted units to be individually connected.

(5) Filter Assembly (A5) - The EMI Filter Assembly shall be hard mounted inside the equipment cabinet at the point of entry for the primary AC power connection to the cabinet.

(6) Video Disc Storage Tray - The Disc Storage Shelf Assembly shall be a standard 19-inch rack mounted assembly secured by front panel fasteners. Stored video discs shall be removable from the front of the equipment without removal of the storage shelf itself.

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(7) Video Disc Recorder Mount - The Video Disc Recorder Mount shall provide standard 19-inch slide mounting for the Video Disc Recorder unit such that the recorder can be extended on the slide rails for maintenance.

(8) SRA Storage Drawer - The SRA Storage Drawer Assembly shall be a slide mounted standard 19-inch rack mounted unit secured by front panel fasteners.

(9) Time Base Corrector (Not Supplied) Mounting - The Time Base Corrector unit, a standard 19-inch rack mounted unit with a nominal 3-1/2 inch panel height, shall be slide mounted and secured by front panel fasteners. (Mounting hardware including slide rails, both chassis and cabinet halves, shall be supplied with the equipment cabinet.)

(10) Video Indicator (Not Supplied) Mounting - The IP-1314/UPQ-5 Video Indicator, a bench mounted unit which is part of the AN/UPQ-5, shall be slide mounted on tiltable slides which allow the unit to be rotated 90 degrees upward or downward when the drawer is extended. The unit shall be secured in the cabinet by front panel fasteners. (Mounting hardware including mounting ears and slide rails, both chassis and cabinet halves, shall be supplied with the equipment cabinet.)

(11) Duplicator Interface Unit (Not Supplied) Mounting - The Duplicator Interface unit, a standard 19-inch rack mounted unit with a nominal 3-1/2 inch panel height, shall be slide mounted and secured by front panel fasteners. (Mounting hardware including slide rails, both chassis and cabinet halves, shall be supplied with the equipment cabinet.)

(12) Frequency Converter (Not Supplied) Mounting - The CV-2993A/UPQ-5 Static Frequency Converter, a standard 19-inch rack mounted unit, shall be slide mounted and secured by front panel fasteners. (Mounting hardware including slide rails, both chassis and cabinet halves, shall be supplied with the equipment cabinet.)

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3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307, and shall apply at the equipment cabinet top assembly only.

3.3.9 Safety - In the design and fabrication of the equipment, the contractor shall conduct a safety program to optimize personnel and equipment safety.

3.3.9.1 Safety Criteria - In the conduct of the safety program, the general requirements of section 5.4.1 of MIL-STD-882A and the following shall be used as a guide:

(1) Section 3.11 of MIL-E-16400G and Amend 1

(2) Section 3.2 of MIL-T-28800B

(3) Section 5.13 of MIL-STD-1472B

3.3.9.2 Order of Precedence - The order of precedence for satisfying system safety criteria and resolving identified hazards shall be as specified in section 5.4.2 of MIL-STD-882A.

3.3.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 95% relative humidity
Input power voltage	115 VAC ± 5%; 60Hz ± 5%

3.3.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental conditions, or reasonable combination of these conditions, as specified herein.

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3.3.11.1 Operating Temperature Range - The equipment shall maintain specified performance when exposed to high and low operating temperatures over the range of 0 to +50 degrees centigrade.

3.3.11.2 Non-Operating Temperature Range - The equipment shall not be damaged nor shall performance be degraded when returned to the operating temperature range after being subjected to non-operating temperatures over the range of -62 to +71 degrees centigrade.

3.3.11.3 Humidity - The equipment shall maintain the specified performance when exposed to a relative humidity of 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment in the form of both water and frost.

3.3.11.4 Salt Fog (Spray) - The equipment shall be capable of withstanding the salt fog test specified in MIL-STD-810, Method 509, Procedure I. Sample corner structures and other critical sections shall be used for the test; the complete equipment shall not be subjected to the test. Samples selected for the salt fog test shall be approved by the procuring activity prior to testing.

3.3.11.5 Shock - The equipment shall be capable of withstanding the shock test specified in MIL-STD-810, Method 516, Procedure I and Figure 516.2.-2 modified as follows: The peak value (A) of the shock wave shall be 10g maximum.

3.3.11.6 Vibration - With the top of the cabinet supported, the equipment shall be capable of meeting the Type I vibration requirements of MIL-STD-167-1 from 5 to 33 Hz except that the vibration in the side-to-side axis shall not exceed 20 Hz.

3.3.12 Input Electrical Power - The equipment shall be supplied with the power cable, terminated with a type WC596/13-3 plug in accordance with W-C-596/13. The equipment shall meet all specified performance requirements when energized from a power source meeting the requirements for Type I power in accordance with DOD-STD-1399/300. The power required shall not exceed the specified amount.

(1) AC Power (Single Phase), 115 V, 60 Hz, Type I, 1150 VA.

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3.3.13 Cooling - Cooling for the equipment shall comply with the requirements of MIL-E-16400 and shall not require the use of specific ducted external cooling. All necessary fans and/or heaters shall be supplied as part of the equipment. With the exception of the space allocated for the IP-1314/UPQ-5 Video Indicator, unused front panel space in the cabinet shall be closed using blank panels to ensure adequate air distribution.

3.3.14 Movability - The equipment shall be designed to permit easy transport.

3.3.14.1 Lifting Eyes - Lifting eyes shall be provided in accordance with MIL-E-16400.

3.3.14.2 Casters - The equipment shall be supplied with four, base mounted, casters for ease of mobility. The casters shall be removable for permanent installations.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall operate in conjunction with the Mission Recorder Display Set AN/UPQ-5 and the Mission Recorder Set AN/USH-17(V) and shall provide power and signal distribution to operational units housed in the rack. The equipment shall also provide overload protection on the input power line to the equipment and shall provide cooling air for operational units housed in the rack.

3.4.1.1 Signal Distribution - The equipment shall provide signal distribution for the composite video signal output from the IP-1101/UPQ-5 Radar Indicator in the AN/UPQ-5 Mission Recorder Display Set. The characteristics of that composite video are as described in MIL-D-81688. The composite video input to the equipment shall be by means of Cable W20 of the AN/UPQ-5. Suitable interface connection shall be provided as part of the equipment described herein. Signal routing cables shall be provided to distribute the composite video through a Time Base Corrector unit to a Video Disc Recorder and from the disc recorder to a Video Indicator IP-1314/UPQ-5. The Time Base Corrector, Video Disc Recorder, and Video Indicator are not part of the equipment covered by this specification but are housed therein.

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Signal routing provisions shall also provide the capability to by-pass either the Time Base Corrector or the Video Disc Recorder. The below listed signal/interconnect cables shall be provided as part of the equipment for functions as indicated.

Cable	Function
W1	Video from Power Panel to Time Base Corrector
W2	Video from Time Base Corrector to Disc Recorder
W3	Video from Disc Recorder to Video Indicator

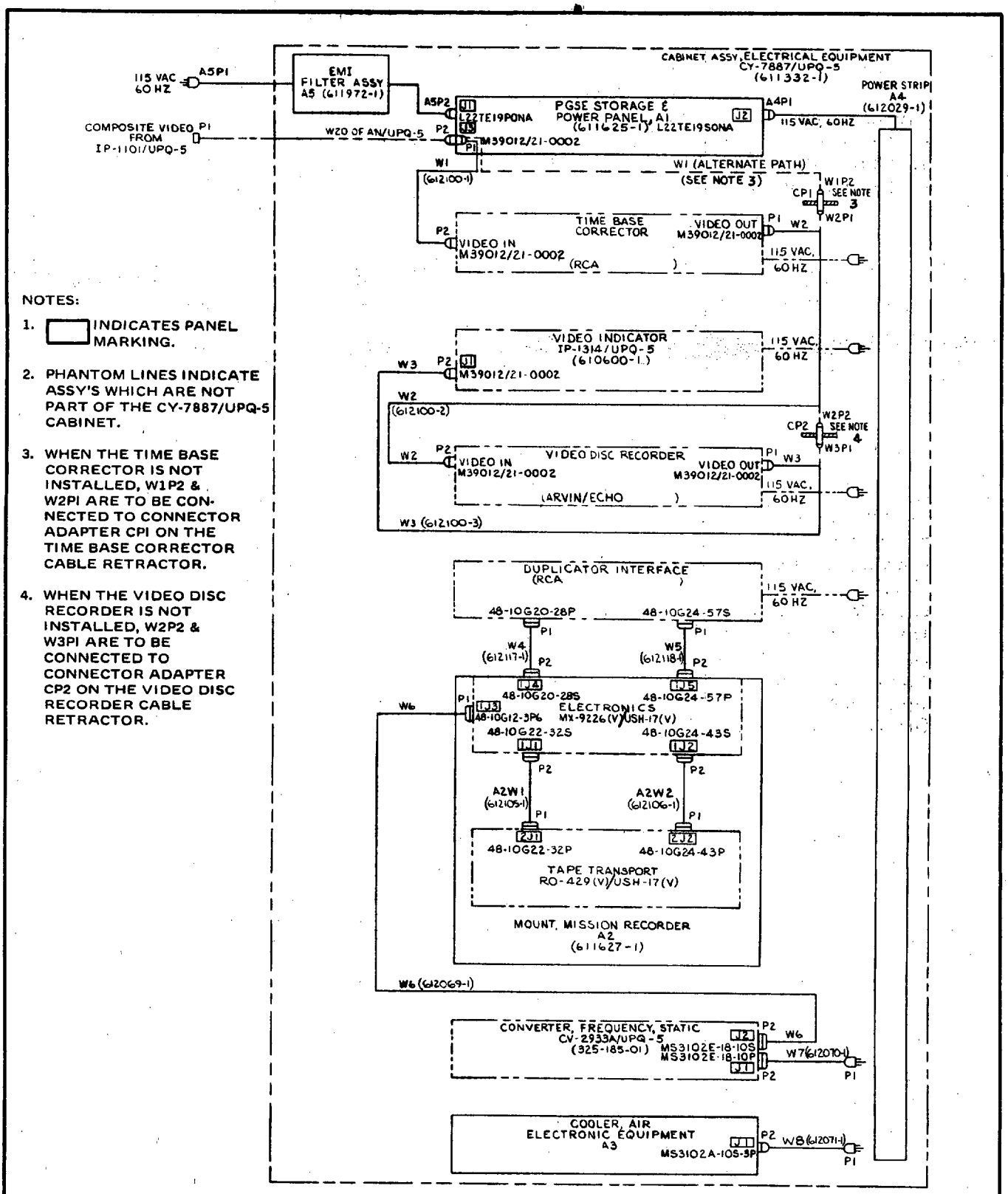
Signal Distribution requirements are illustrated in Figure 1.

3.4.1.2 Power Distribution - The equipment shall provide distribution of 10 amps of EMI-filtered, 115 volt, 60 Hz, single-phase power from a single input power cord to a minimum of nine individual three-prong standard AC receptacles. Each individual receptacle shall be capable of providing up to 10 amps of AC current, but the total current supplied by all individual outlets shall not exceed 10 amps. The maximum voltage drop between the primary source voltage to which the equipment is connected and the measured voltage at any internal receptacle shall not exceed 0.5 volt AC. The below listed power/interconnect cables shall be provided as part of the equipment for functions as indicated.

Cable	Function
W4	Interconnect AN/USH-17(V) Electronics Package to Duplicator Interface Unit
W5	Interconnect AN/USH-17(V) Electronics Package to Duplicator Interface Unit
W6	400 Hz Power to Tape Duplicator
W7	60 Hz Power to Freq. Converter
W8	60 Hz Power to Air Cooler
A2W1	Interconnect AN/USH-17(V) Electronics Package and Transport Assy
A2W2	Interconnect AN/USH-17(V) Electronics Package and Transport Assy

Power distribution requirements are illustrated in Figure 1.

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NOTES:

1. ▭ INDICATES PANEL MARKING.
2. PHANTOM LINES INDICATE ASSY'S WHICH ARE NOT PART OF THE CY-7887/UPQ-5 CABINET.
3. WHEN THE TIME BASE CORRECTOR IS NOT INSTALLED, W1P2 & W2P1 ARE TO BE CONNECTED TO CONNECTOR ADAPTER CP1 ON THE TIME BASE CORRECTOR CABLE RETRACTOR.
4. WHEN THE VIDEO DISC RECORDER IS NOT INSTALLED, W2P2 & W3P1 ARE TO BE CONNECTED TO CONNECTOR ADAPTER CP2 ON THE VIDEO DISC RECORDER CABLE RETRACTOR.

FIG. 1 CABLING DIAGRAM

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3.4.1.3 Short-Circuit Protection - The equipment shall provide circuit breaker protection such that a short-circuit condition anywhere in the filtered 60 Hz power distribution network interrupts the AC power.

3.4.1.4 Cooling Air - The equipment shall provide cooling air for equipment mounted therein. Air shall be drawn into the bottom of the equipment cabinet by a blower and exhaust louvers shall be provided in the side panels near the top of the cabinet. Air volume of the blower shall be as specified in Section 3.5.

3.5 Detailed Requirements -

3.5.1 Cabinet (Subassembly) - The cabinet subassembly shall meet the following requirements:

3.5.1.1 Function - The cabinet subassembly shall provide an enclosure, mounting facilities, and interconnecting cabling for the equipments/assemblies listed below.

(1) PGSE Storage & Power Panel Assembly (A1)
(Part of the equipment supplied hereunder.)

(2) Disc Storage Shelf Assembly
(Part of the equipment supplied hereunder.)

(3) Time Base Corrector
(Proposed unit; not part of the equipment delivered hereunder.)

(4) Video Indicator IP-1314/UPQ-5
(Part of the AN/UPQ-5 Mission Recorder Set; not part of the equipment supplied hereunder.)

(5) Video Disc Recorder
(Proposed unit; not part of the equipment delivered hereunder.)

(6) Duplicator Interface
(Proposed unit, not part of the equipment delivered hereunder.)

(7) Mission Recorder Mount (A2)
(Part of the equipment delivered hereunder.)

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(8) CV-2933A/UPQ-5 Static Frequency Converter
(Existing item of supply; not part of the equipment supplied hereunder.)

(9) SRA Storage Drawer Assembly
(Part of the equipment delivered hereunder.)

(10) Electronic Equipment Air Cooler (A3)
(Part of the equipment delivered hereunder.)

3.5.1.2 Form Factor - The cabinet shall be a ruggedized, low-carbon steel, EMI shielded unit and shall not exceed 24 inches in width, 27 inches in depth, and 72 inches in height. The cabinet shall, to the extent feasible, be similar to the Electrical Equipment Cabinet CY-7913A/UPQ-5 as described in MIL-D-81688 for the AN/UPQ-5.

3.5.1.3 EMI Gasketing - EMI gasketing shall be provided for both front and rear access panels of the cabinet.

3.5.1.4 Mounting Rails - Mounting rails shall be provided in the cabinet to accommodate standard 19-inch, front panel mounted units. Mounting rails shall provide EIA standard hole spacing.

3.5.1.5 Weight - The total weight of the cabinet, including cables, slides, and other components thereof but excluding the rack mounted assemblies listed in section 3.5.1.1, shall not exceed 500 pounds.

3.5.1.6 Contents - The cabinet subassembly shall contain the following.

(1) Time Base Corrector Mounting - Facilities shall be provided to slide mount a Time Base Corrector (TBC). The TBC is a standard 19-inch, rack mounted unit having a nominal panel height of 3-1/2 inches. The slide mounts shall allow the TBC unit to be fully extended from the cabinet for maintenance activities and shall incorporate a means to lock the slides in the extended position. Both the cabinet and chassis mounted portion of the slide mechanism shall be supplied with the cabinet. Cable retractors shall be provided to allow cables connected to the rear of the TBC to be safely and automatically withdrawn as the drawer is closed.

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(2) IP-1314/UPQ-5 Video Indicator Mounting - Facilities shall be provided to slide mount an IP-1314/UPQ-5 Video Indicator on tiltable slides which allow the unit to be fully extended from the cabinet and rotated 90 degrees upward or downward for maintenance activities. A means shall be provided to lock the slides in the extended position. The IP-1314/UPQ-5 is a bench mounted unit measuring 8-3/4 inches high, 16-5/8 inches wide, and 22 inches deep excluding handles and control knobs. The IP-1314/UPQ-5 is defined in MIL-D-81688. Both the cabinet and chassis mounted portion of the slide mechanism shall be supplied with the cabinet. The mounting brackets required to adapt the IP-1314 for rack mounting shall be supplied as part of the cabinet addressed herein. Cable retractors shall be provided to allow cables connected to the rear of the IP-1314 to be safely and automatically withdrawn as the drawer is closed.

(3) Video Disc Recorder (VDR) Mounting - A Rack Mounting Kit, Part Number 335-804-01 (FSMC 99592) shall be provided as part of the equipment cabinet to facilitate rack mounting of the VDR. Cable retractors shall be provided to allow cables attached to the rear of the VDR to be safely and automatically withdrawn as the drawer is closed.

(4) Duplicator Interface Mounting - Facilities shall be provided to slide mount a Duplicator Interface unit. The Duplicator Interface unit is a standard 19-inch rack mounted unit with a nominal panel height of 3-1/2 inches. The slide mounts shall allow the Duplicator Interface unit to be fully extended from the cabinet for maintenance activities and shall incorporate a means to lock the slides in the extended position. Both the cabinet and chassis mounted portion of the slide mechanism shall be supplied with the cabinet. Cable retractors shall be provided to allow cables connected to the rear of the Duplicator Interface to be safely and automatically withdrawn as the drawer is closed.

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(5) Mission Recorder Mount Mounting (A2) - Facilities shall be provided to slide mount the Mission Recorder Mount. (The Mission Recorder Mount is a subassembly of and is deliverable as part of the Electrical Equipment Cabinet CY-7887/UPQ-5.) The slide mount shall allow the Mission Recorder Mount to be fully extended from the cabinet for maintenance and operational activities and shall incorporate a means to lock the slides in the extended position. Only the cabinet mounted portion of the slide mechanism shall be part of the cabinet subassembly; the chassis mounted portion shall be part of the Mission Recorder Mount addressed in section 3.5.4 of this specification. Cable retractors shall be provided to allow cables connected to the rear of the Mission Recorder Mount to be safely and automatically withdrawn as the drawer is closed.

(6) CV-2933A/UPQ-5 Static Frequency Converter Mounting - Facilities shall be provided to slide mount a CV-2933A/UPQ-5. The CV-2933A/UPQ-5 is a rack mounted unit measuring 3-1/2 inches high, 19 inches wide, and 19 inches deep and is defined in MIL-D-81688. The slides shall allow the CV-2933A/UPQ-5 to be fully extended from the cabinet for maintenance activities and shall incorporate a means to lock the slides in the extended position. Both the cabinet and chassis mounted portion of the slides shall be supplied with the cabinet. Cable retractors shall be provided to allow cables connected to the rear of the CV-2933A/UPQ-5 to be safely and automatically withdrawn as the drawer is closed.

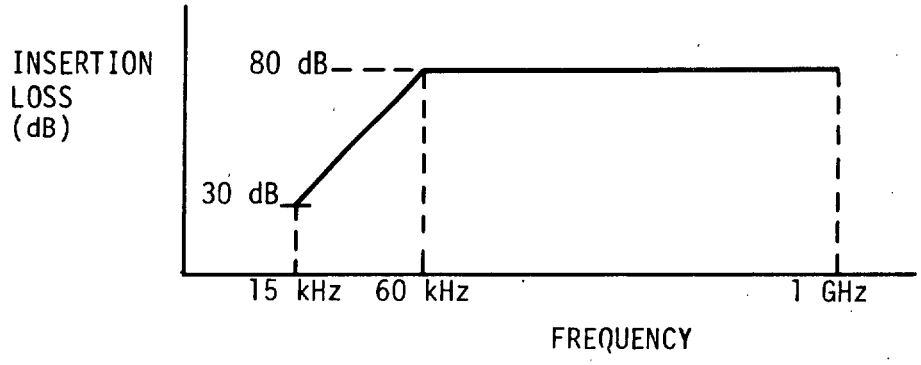
(7) SRA Storage Drawer Mounting - Facilities shall be provided to slide mount the SRA Storage Drawer Assembly. (The storage drawer is a subassembly of and is deliverable as part of the Electrical Equipment Cabinet CY-7887/UPQ-5.)

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The slide mount shall allow the storage drawer to be fully extended from the cabinet to facilitate removal of the spare SRA's stored therein and shall provide a means to lock the slides in the extended position. Only the cabinet mounted portion of the slide mechanism shall be part of the cabinet subassembly; the chassis mounted portion shall be part of the Storage Drawer Assembly addressed in section 3.5.5 of this specification.

(8) Power Strip (A4) - An AC power strip shall be provided to distribute 115 V, 60 Hz, single-phase power from the Power Panel (Part of A1) to a minimum of nine standard, three-prong AC receptacles. The power strip shall provide at least 10-amp total service with each individual receptacle capable of supplying any part of that service up to and including 10 amps.

(9) EMI Filter Assembly (A5) - The EMI Filter Assembly shall provide EMI filtering of the input power line ahead of the Power Panel (Part of A1). The EMI filter shall be comprised of two double-L, 10 amp line filters with a minimum insertion loss at rated load of 30 dB at 15 KHz. The insertion loss shall increase with frequency such that the minimum insertion loss is 80 dB for frequencies from 60 KHz to 1 GHz as shown below:



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The input power cord and the cable to the Power Panel shall be part of the EMI Filter Assembly and shall employ connectors as shown below:

Reference Designation	Receptacle Type	Function
A5P1	WC596/13-3	Primary power input
A5P2	L22TF19S6NA (180-592-01, FSCM 99592)	Filtered power output

(10) Cable Assemblies - The cabinet subassembly shall include the cable assemblies listed below to interconnect the assemblies/units housed in the CY-7887/UPQ-5 Electrical Equipment Cabinet. The cable assemblies shall employ connector types as shown. (Refer to Figure 1 for the Cabling Diagram)

Cable Ref. Des.	Connector Ref. Des.	Receptacle Type	Function
W1	W1P1 W1P2	M39012/16-0101 M39012/16-0101	Composite Video from Power Panel to TBC input or CP1
W2	W2P1 W2P2	M39012/16-0101 M39012/16-0101	Composite Video from TBC output or CP1 to VDR input or CP2
W3	W3P1 W3P2	M39012/16-0101 M39012/16-0101	Composite Video from VDR output or CP2 to IP-1314/UPQ-5 input
W4	W4P1 W4P2	MS24266G20B28SN MS24266G20B28PN	Interconnection between MX-9226(V)/USH-17(V), J4 and the Duplicator Interface Unit
W5	W5P1 W5P2	MS24266G24B57PN MS24266G24B57SN	Interconnection between MX-9226(V)/USH-17(V), J5 and the Duplicator Interface Unit

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Cable Ref. Des.	Connector Ref. Des.	Receptacle Type	Function
W6	W6P1 W6P2	MS24266R12B3S6X MS3106A-18-10P	400 Hz power input to MX-9226(V)/USH-17(V), J3 from CV-2933A/UPQ-5, J2
W7	W7P1 W7P2	UP-221M MS3106A-18-10S	60 Hz power input to CV-2933A/UPQ-5, J1
W8	W8P1 W8P2	UP-221M MS3106A-10SL-3S	60 Hz power input to Cooler

(11) BNC Adapters - Two BNC feedthrough type adapters (CP1 and CP2) in accordance with MIL-A-55339/13 (P/N M55339/13-00492) shall be provided to facilitate signal bypass of either the Time Base Corrector or the Video Disc Recorder.

3.5.2 PGSE Storage & Power Panel Assembly (A1) - The PGSE Storage and Power Panel Assembly (PSPP) shall meet the following requirements:

3.5.2.1 Function - The PSPP shall provide non-operating storage facilities for two items of Peculiar Ground Support Equipment (PGSE) associated with the AN/UPQ-5 Mission Recorder Display Set. The two items of PGSE are the SM-697/UPQ-5 Radar Signal Simulator described in MIL-S-81977 and the DA-662/UPQ-5 Electrical Dummy Load described in MIL-D-81976. The PSPP shall also provide the means to control (ON/OFF and short circuit protection) for the 115 VAC, 60 Hz input power to the CY-7887/UPQ-5. Additionally, the PSPP shall provide the primary signal interface for the Composite Video from the AN/UPQ-5 to the CY-7887/UPQ-5.

3.5.2.2 Form Factor - The PSPP shall be a standard 19-inch rack mounted unit and shall not exceed 10-1/2 inches high, 19 inches wide, and 19-1/4 inches deep not including knobs, connectors, and front panel hardware.

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3.5.2.3 Weight - The weight of the PGSE Storage and Power Panel Assembly shall not exceed 20 pounds.

3.5.2.4 Controls - The power panel shall have a single control which provides both the ON/OFF and short circuit protection, i.e., a circuit breaker. The circuit breaker shall be a type M39019/5-74 conforming to the requirements of MIL-C-39019/5.

3.5.2.5 Indicators - The power panel shall provide an indicator which shall be lighted when the circuit breaker is set to the ON position and 115 VAC is actively applied to the CY-7887/UPQ-5.

3.5.2.6 Electrical Connectors - Connections to external circuits shall be provided as follows (see Figure 1):

Reference Designation	Receptacle Type	Function
J1	L22TE19PONA (180-591-01, FSCM 99592)	EMI-Filtered power input
J2	L22TE19SONA (180-591-02, FSCM 99592)	Switched power output
J3	M55339/13-00492 (MIL-A-55339/13)	Composite Video input

3.5.2.7 PGSE Storage Facilities - The PGSE storage facilities shall accommodate non-operating storage of one SM-697/UPQ-5 Radar Signal Simulator and one DA-662/UPQ-5 Electrical Dummy Load. Fasteners/hold-downs shall be provided to secure the items when stored; these fasteners/hold-downs shall permit easy removal of either item for use.

3.5.3 Disc Storage Shelf Assembly - The Disc Storage Shelf Assembly shall meet the following requirements:

3.5.3.1 Function - The Disc Storage Shelf Assembly shall provide storage facilities for a minimum of two magnetic video recording discs, each housed in cassette-type enclosure. Each recording disc measures 13.5 inches by 13.75 inches by 0.375 inches and weighs 28 ounces.

3.5.3.2 Form Factor - The Disc Storage Shelf Assembly shall be a standard 19-inch rack mounted unit not exceeding 1-3/4 inches high, 19 inches wide, and 22-1/2 inches deep.

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3.5.3.3 Weight - The weight of the Disc Storage Shelf shall not exceed 6 pounds.

3.5.4 Mission Recorder Mount (A2) - The Mission Recorder Mount shall meet the following requirements:

3.5.4.1 Function - The Mission Recorder Mount shall support, position and provide electrical interconnection between the MX-9226(V)/USH-17(V) Electronics and the RO-429(V)/USH-17(V) Tape Transport components of a uniquely configured AN/USH-17(V) Mission Recorder Set.

3.5.4.2 Form Factor - The Mission Recorder Mount shall be a standard 19-inch rack mounted unit not exceeding 10-1/2 inches high, 19 inches wide, and 24-1/2 inches deep excluding front panel fasteners.

3.5.4.2.1 Operational Access - The Tape Transport shall be positioned such that the tape can be changed without disconnecting the unit from the mount.

3.5.4.3 Weight - The weight of the Mission Recorder Mount shall not exceed 34 pounds.

3.5.4.4 Contents - The Mission Recorder Mount shall include the cable assemblies listed below. The cable assemblies shall employ connector types as shown. (Refer to Figure 1 for the Cabling Diagram.)

Cable Ref. Des.	Connector Ref. Des.	Receptacle Type	Function
A2W1	A2W1P1	48-16G22-32S (180-596-02, FSCM 99592)	Interconnection between MX-9226(V)/USH-17(V), J1 and RO-429(V)/USH-17(V) J1
	A2W1P2	48-16G22-32P (180-596-01, FSCM 99592)	
A2W2	A2W2P1	48-16G24-43S (180-596-04, FSCM 99592)	Interconnection between MX-9226(V)/USH-17(V), J2 and RO-429(V)/USH-17(V) J2
	A2W2P2	48-16G24-43P (180-596-03, FSCM 99592)	

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3.5.5 SRA Storage Drawer Assembly - The SRA Storage Drawer Assembly shall meet the following requirements:

3.5.5.1 Function - The SRA Storage Drawer Assembly shall provide non-operating storage for selected spare Shop Replaceable Assemblies (SRA's) of the AN/UPQ-5 Mission Recorder Display Set and the AN/USH-17(V) Mission Recorder Set. The SRA's to be stored are listed in Table I.

3.5.5.2 Form Factor - The SRA Storage Drawer Assembly shall be a standard 19-inch rack mounted drawer assembly not exceeding 8-3/4 inches high, 19 inches wide, and 24-1/4 inches deep excluding handles.

3.5.5.2.1 Accessibility - Each stored SRA shall be removable without the necessity to first remove any other stored SRA.

3.5.5.2.2 SRA Retention - Stored SRA's shall be mounted within the drawer such that they are retained in their proper position during service conditions cited elsewhere in this specification. Loose storage of SRA's shall not be used.

3.5.5.3 Weight - The weight of the SRA Storage Drawer Assembly shall not exceed 24 pounds.

3.5.5.4 Marking - The reference designation for each of the spare SRA's listed in Table 1 shall be marked adjacent to the storage location allocated for the particular SRA.

3.5.6 Electronic Equipment Air Cooler (A3) - The Electronic Equipment Air Cooler shall meet the following requirements:

3.5.6.1 Function - The air cooler shall draw ambient air into the bottom of the cabinet, filter it, and circulate it upward to provide cooling for equipment mounted in the cabinet.

3.5.6.2 Form Factor - The air cooler shall be a standard 19-inch rack mounted, EMI shield unit and shall not exceed 5-1/4 inches panel height and 6-1/8 inches high overall, 19 inches wide, and 13 inches deep.

3.5.6.3 Weight - The weight of the air cooler shall not exceed 20 pounds.

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TABLE I STORED SRA'S

ITEM NO.	REF. DES.	DESCRIPTION	PART NO.	FSCM
		RADAR INDICATOR MODULES		
1	A1A4 A4A4	HV Power Supply Assy	601221-1	99592
2	A1A7 A4A7	Circuit Card Assy, Video Amplifier	603566-1	99592
3	A1A8	Circuit Card Assy, Local-Remote Control	603563-1	99592
4	A1A9 A4A9	Circuit Card Assy, Unblanking and Phosphor Protect	601194-1	99592
5	A1A10	Circuit Card Assy, Audio Amplifier	601190-1	99592
6	A1A11	Circuit Card Assy, Shutter Sync Horizontal Sweep Sampler	603572-1	99592
7	A1A12	Circuit Card Assy, Shutter Sync Edge Detector and Logic	603570-1	99592
8	A4A12	Circuit Card Assy, Test Signal Generator	601186-1	99592
9	A1A13 A4A13	Circuit Card Assy, Fault Sensing	601196-1	99592
10	A1A15 A4A15	Bias Supply	601313-1	99592
11	A1A5 A4A5 A1A6 A4A6	Amplifier Driver, Deflection Coil	601222-1	99592

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TABLE I STORED SRA'S (CONT'D)

ITEM NO.	REF. DES.	DESCRIPTION	PART NO.	FSCM
		DATA DISPLAY MODULES		
12	A2A2 A2A3 A2A4	Circuit Card Assy, Two-Digit Data Freeze Memory	600381-1	99592
13	A2A5	Circuit Card Assy, BCD Counter	600400-1	99592
14	A2A6	Circuit Card Assy, Scaled Clock Control	604185-1	99592
15	A2A7 A2A8 A2A9	Circuit Card Assy, Two-Digit Data Block Memory	600404-1	99592
16	A2A10	Circuit Card Assy, Clock Generator	600345-1	99592
17	A2A11	Circuit Card Assy, Data Freeze Control "C"	600420-1	99592
18	A2A12	Circuit Card Assy, Data Freeze Control "A"	600415-1	99592
19	A2A13	Circuit Card Assy, Sync Detector and Decoder	600341-1	99592
20	A2A14	Circuit Card Assy, Shutter Control and Sign Storage	600429-2	99592
21	A2A15 A2A16 A2A17	Circuit Card Assy, Readout Drivers	600417-1	99592
22	A2A18	Circuit Card Assy, Data Freeze Control "B"	600411-1	99592
23	A2A19	Circuit Card Assy, Freeze Address Decoder and Comparator	604217-1	99592

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TABLE I STORED SRA'S (CONT'D)

ITEM NO.	REF. DES.	DESCRIPTION	PART NO.	FSCM
24	A2A20	Circuit Card Assy, Stored Scale Factor "A"	604182-1	99592
25	A2A21	Circuit Card Assy, Data and Address Storage	600369-1	99592
26	A2A22	Circuit Card Assy, Selected Scale Factors	600499-1	99592
27	A2A23	Circuit Card Assy, Binary Counter and Fixed Scale Factors	600335-1	99592
28	A2A24	Circuit Card Assy, Data Storage and Transition Gates	600363-1	99592
29	A2A25	Circuit Card Assy, Block Address Decoder and Comparator	600354-1	99592
30	A2A26	Circuit Card Assy, Data Block Control	600406-2	99592
31	A2A27	Circuit Card Assy, Stored Scale Factor "B"	604183-1	99592
32	A2A28	Circuit Card Assy, Error Sensing	600440-1	99592
33	A2A29	Circuit Card Assy, Self-Test	600402-2	99592
		TRAM DISPLAY MODULES		
34	A2	HV Power Supply Assy	610610-1	99592
35	A3	Circuit Card Assy, Line Rate Deflection Amplifier	610613-1	99592
36	A4	Circuit Card Assy, Field Rate Deflection Amplifier	610616-1	99592

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TABLE I STORED SRA'S (CONT'D)

ITEM NO.	REF. DES.	DESCRIPTION	PART NO.	FSCM
37	A5	Circuit Card Assy, Sync Processor	610619-1	99592
38	A6	Circuit Card Assy, Sync Detector	610622-1	99592
39	A7	Circuit Card Assy, Video Amplifier	610625-1	99592
40	A8	Circuit Card Assy, BIT Generator	610603-1	99592
41	A9	Circuit Card Assy, Unblanking and Phosphor Protect	601194-1	99592
		MISSION RECORDER MODULES		
42	---	Circuit Card Assy, Module No. 1	TBD 1/	49671
43	---	Circuit Card Assy, Module No. 2	TBD 1/	49671

1/ Modules No. 1 and No. 2 are Circuit Card Assemblies 5-5/8 inches long, 4 inches wide, with a board thickness of 1/16 inch and module thickness (including components) of 5/8 inch.

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3.5.6.4 Capacity - The air cooler shall provide up to 290 cubic feet per minute of cooling air.

3.5.6.5 Power Requirements - The air cooler shall operate on 115 volts, single phase, 50/60 Hz at a nominal running current of 1.3 amps.

3.5.6.6 Electrical Connectors - Connection to external circuits shall be provided as follows (See Figure 1):

Reference Designation	Receptacle Type	Function
J1	MS3102A-10S-3P	115 VAC input power

3.5.7 Blank Panels - Blank front panels supplied with the equipment shall meet the following requirements:

3.5.7.1 Function - The blank front panels shall close unused front panel space in the cabinet to ensure adequate distribution of cooling air. (These panels are removed when the resident units are installed in the cabinet.)

3.5.7.2 Form Factor - Blank panels shall be compatible with EIA standard 19-inch rack mounting and shall meet the following requirements:

3.5.7.2.1 Panel (Small) - The small pattern panels provided to close off the Top, Time Base Corrector, Duplicator Interface, and Frequency Converter rack space shall be 3-15/32 inches high, 19 inches wide, and 0.125 inch thick.

3.5.7.2.2 Panel (Large) - The large pattern panel provided to close off the Video Disc Recorder rack space shall be 6-31/32 inches high, 19 inches wide, and 0.125 inch thick.

3.5.7.2.3 Panel (Special) - The special panel provided to close off the front of the Mission Recorder Mount shall be 10-15/32 inches high, 19 inches wide and 0.062 inch thick with a 5/16 inch lip on the top edge.

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3.5.8 Interface Requirements - The following is a compilation of the special interface requirements for units to be mounted in the equipment cabinet described in this specification.

3.5.8.1 SM-697/UPQ-5 Interface Requirements - The interface considerations for the Radar Signal Simulator, SM-697/UPQ-5, are as follows:

(1) Mechanical - The SM-697/UPQ-5 is a portable unit 4-3/8 inches high, 13-1/8 inches wide, and 17-1/2 inches deep weighing 27 pounds.

(2) Electrical - There are no electrical interface requirements for the SM-697/UPQ-5 in association with this specification. (Only non-operational storage is provided.)

(3) Reference Documents - Additional interface data may be found by reference to the following documents:

(a) Military Specification - MIL-S-81977

(b) Engineering Drawing Package - Dwg. No. 603680 (FSCM 99592) along with subordinate drawing and associated lists.

(c) Technical Manual - NAVAIR 16-35SM697-1

3.5.8.2 DA-662/UPQ-5 Interface Requirements - The interface requirements for the Electrical Dummy Load, DA-662/UPQ-5 are as follows:

(1) Mechanical - The DA-662/UPQ-5 is a portable unit 4-3/8 inches high, 13-1/8 inches wide, and 18 inches deep weighing 23 pounds.

(2) Electrical - There are no electrical interface requirements for the DA-662/UPQ-5 in association with this specification.(Only non-operational storage is provided.)

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(3) Reference Documents - Additional interface data may be found by reference to the following documents:

(a) Military Specification - MIL-D-81976

(b) Engineering Drawing Package - Dwg. No. 603685 (FSCM 99592) along with subordinate drawings and associated lists.

(c) Technical Manual - NAVAIR 16-35SM697-1

3.5.8.3 Time Base Corrector Interface Requirements - The interface requirements for the Time Base Corrector (TBC) are as follows:

(1) Mechanical - The Time Base Corrector, when developed, will be a standard 19-inch rack mounted unit with a nominal panel height of 3-1/2 inches, a behind the panel width of 16-7/8 inches, and a behind the panel depth of 12 inches minimum and 17 inches maximum. The maximum weight of the TBC will not exceed 40 pounds.

(2) Electrical - The 115VAC power is applied to the TBC through a power cord integral to the unit. Signal interface to the TBC is accomplished by Cable Assemblies W1 and W2. (See Figure 1, and sections 3.5.1 and 3.5.8.10)

3.5.8.4 IP-1314/UPQ-5 Interface Requirements - The interface requirements for the Video Indicator, IP-1314/UPQ-5 are as follows:

(1) Mechanical - The IP-1314/UPQ-5 is a portable, bench mounted unit 8-3/4 inches high, 16-5/8 inches wide and 22 inches deep weighing 65 pounds.

(2) Electrical - There are two electrical interface requirements for the Video Indicator: 115 VAC power input, and signal input. The power interface is accomplished by means of the line cord integral to the Video Indicator. The signal interface to the Video Indicator is by means of cable W3. (See sections 3.5.1 and 3.5.8.10.)

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(3) Reference Documents - Additional interface data may be found by reference to the following documents:

(a) Military Specification - MIL-D-81688

(b) Engineering Drawing Package - Dwg. No. 610600 (FSCM 99592) along with subordinate drawings and associated lists.

(c) Technical Manual - NAVAIR 16-30UPQ5-1 and NAVAIR 16-30UPQ5-4

3.5.8.5 Video Disc Recorder Interface Requirements - The interface requirements for the Video Disc Recorder (VDR) are as follows:

(1) Mechanical - The Video Disc Recorder, when developed, will be a portable unit 7 inches high, 16-1/2 inches wide, and 17-1/2 inches deep. The maximum weight shall not exceed 40 pounds. The mounting configuration will be compatible with the rack mounting kit specified in section 3.5.1.

(2) Electrical - The 115 VAC power is applied to the VDR through a power cord integral to the unit. Signal interface is by Cable Assemblies W2 and W3. (See Figure 1, and sections 3.5.1 and 3.5.8.10.)

3.5.8.6 Duplicator Interface Unit Interface Requirements - The interface requirements for the Duplicator Interface Unit (DI/F) are as follows:

(1) Mechanical - The Duplicator Interface unit, when developed, will be standard 19-inch rack mounted unit with a nominal panel height of 3-1/2 inches, a behind the panel width of 16-7/8 inches, and a behind the panel depth of 12 inches minimum and 17 inches maximum. The maximum weight of the DI/F will not exceed 40 pounds.

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(2) Electrical - The 115 VAC power is applied to the DI/F through a power cord integral to the unit. The electrical interface (power/signal) between the DI/F and the Electronics package (MX-9226(V)/USH-17(V)) of a uniquely configured AN/USH-17(V) Mission Recorder Set is provided by Cable Assemblies W4 and W5 (See Figure 1, and sections 3.5.1 and 3.5.8.10.)

(3) Reference Documents - Additional interface data may be found by reference to the following documents:

(a) Military Specification - MIL-R-81687

(b) Engineering Drawing Package - Dwg. No. 8370986 (FSCM 49671) along with subordinate drawings and associated lists.

(c) Technical Manual - NAVAIR 16-30USH17-1 and NAVAIR 16-30USH17-2

NOTE

Suppliers are cautioned the above documents may, or may not, address the unique configuration of the AN/USH-17(V) employed as the Tape Duplicator.

3.5.8.7 AN/USH-17(V) Interface Requirements - The two primary components of the AN/USH-17(V) Mission Recorder Set which have direct interface relationship to equipment cabinet described herein are the MX-9226(V)/USH-17(V) Electronics and the RO-429(V)/USH-17(V) Transport. The interface requirements for these components are as follows:

(1) Mechanical -

(a) Electronics - The Electronics unit is enclosed in a case nominally 4-1/2 inches high, 12 inches wide, and 23-1/2 inches long, exclusive of controls, handles, and mounting tabs. The unit weighs approximately 45 pounds.

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(b) Transport - The Transport unit is enclosed in a case nominally 4 inches high, 12 inches wide, and 23-1/2 inches long, exclusive of connectors and handles. The unit weighs approximately 45 pounds.

(2) Electrical -

(a) Electronics Unit to Duplicator Interface Unit - The electrical interface between the MX-9226(V)/USH-17(V) Electronics unit and the Duplicator Interface unit is accomplished by Cable Assemblies W4 and W5. (See Figure 1, and sections 3.5.1 and 3.5.8.10.)

(b) Electronics Unit to Static Frequency Converter - The 400 Hz input power interface from the CV-2933A/UPQ-5 Static Frequency Converter to the Electronics unit is accomplished by Cable Assembly W6. (See Figure 1, and sections 3.5.1 and 3.5.8.10.)

(c) Electronics Unit to Transport Unit - The electrical interface between the Electronics unit and the transport unit is provided by Cable Assemblies A2W1 and A2W2, which are part of the Mission Recorder Mount supplied with the equipment cabinet described in this specification. (See Figure 1, and sections 3.5.4 and 3.5.8.10.)

(3) Reference Documents - Additional interface data may be found by reference to the following documents:

(a) Military Specification - MIL-R-81687

(b) Engineering Drawing Package - Dwg No. 8370986 (FSCM 49671) along with subordinate drawings and associated lists.

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(c) Technical Manual - NAVAIR 16-30USH17-1 and NAVAIR 16-30USH17-2

NOTE

Suppliers are cautioned the above documents may, or may not, address the unique configuration of the AN/USH-17(V) employed as the Tape Duplicator.

3.5.8.8 CV-2933A/UPQ-5 Interface Requirements - The CV-2933A/UPQ-5 Static Frequency Converter is the same unit used in the AN/UPQ-5 Mission Recorder Display Set. The interface requirements for the CV-2933A/UPQ-5 are as follows:

(1) Mechanical - The CV-2933A/UPQ-5 is a standard 19-inch rack mounted unit nominally 3-1/2 inches high, 19 inches wide and 19 inches deep. The unit weighs approximately 65 pounds.

(2) Electrical - The 115 VAC, 60 Hz power input is provided to the CV-2933A/UPQ-5 by Cable Assembly W7. The 115 VAC, 400 Hz output from the CV-2933A/UPQ-5 is accommodated by Cable Assembly W6 (See Figure 1, and sections 3.5.1 and 3.5.8.10.)

(3) Reference Documents - Additional interface data may be found by reference to the following documents:

(a) Military Specification - MIL-D-81688

(b) Engineering Drawing Package - Dwg. No. 325-185 (FSCM 99592)

(c) Technical Manual - NAVAIR 16-30UPQ5-1

3.5.8.9 Spare SRA Interface Requirements - The interface requirements for the spare SRA's are as follows:

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(1) Mechanical - The spare SRA's to be stored (non-operating) are selected modular assemblies and circuit card assemblies from the AN/UPQ-5 Mission Recorder Set and two circuit card assemblies to be developed for the uniquely configured AN/USH-17(V) Mission Recorder Set used herewith. Individual SRA's are listed and identified by part number and FSCM in Table I of this specification.

(2) Electrical - There is no electrical interface required for the stored spare SRA's, they are stored in a non-operating mode.

(3) Reference Documents - Additional interface data for the AN/UPQ-5 spare SRA's may be found by reference to the following documents:

(a) Military Specification - MIL-D-81688

(b) Engineering Drawing Package - Dwg No. 601209 (FSCM 99592) along with subordinate drawings and associated lists.

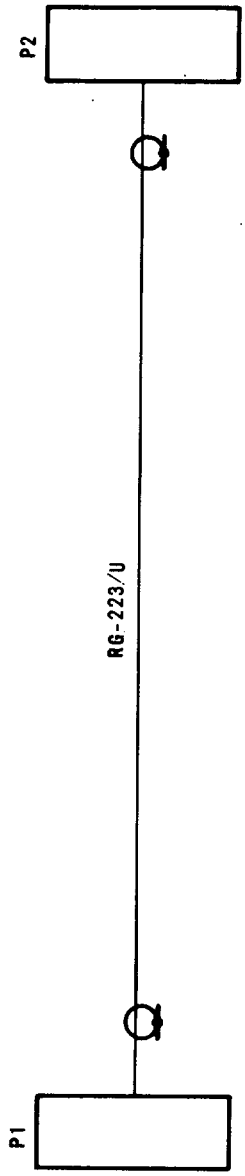
(c) Technical Manual - NAVAIR 16-30UPQ5-1

3.5.8.10 Cable Assembly Interface Requirements - The connector to connector interface requirements for Cable Assemblies W1 through W8, A2W1, and A2W2 are shown in Figures 2 through 9. Table II presents a cross reference between the Cable Assembly reference designators and the applicable figure.

TABLE II CABLE INTERFACE

REF. DES.	FIGURE
W1, W2, & W3	Fig. 2
W4	Fig. 3
W5	Fig. 4
W6	Fig. 5
W7	Fig. 6
W8	Fig. 7
A2W1	Fig. 8
A2W2	Fig. 9

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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.1.6(10)

FIG. 2 INTERFACE REQUIREMENTS, CABLE ASSY W1, W2, & W3

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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.1.6(10)

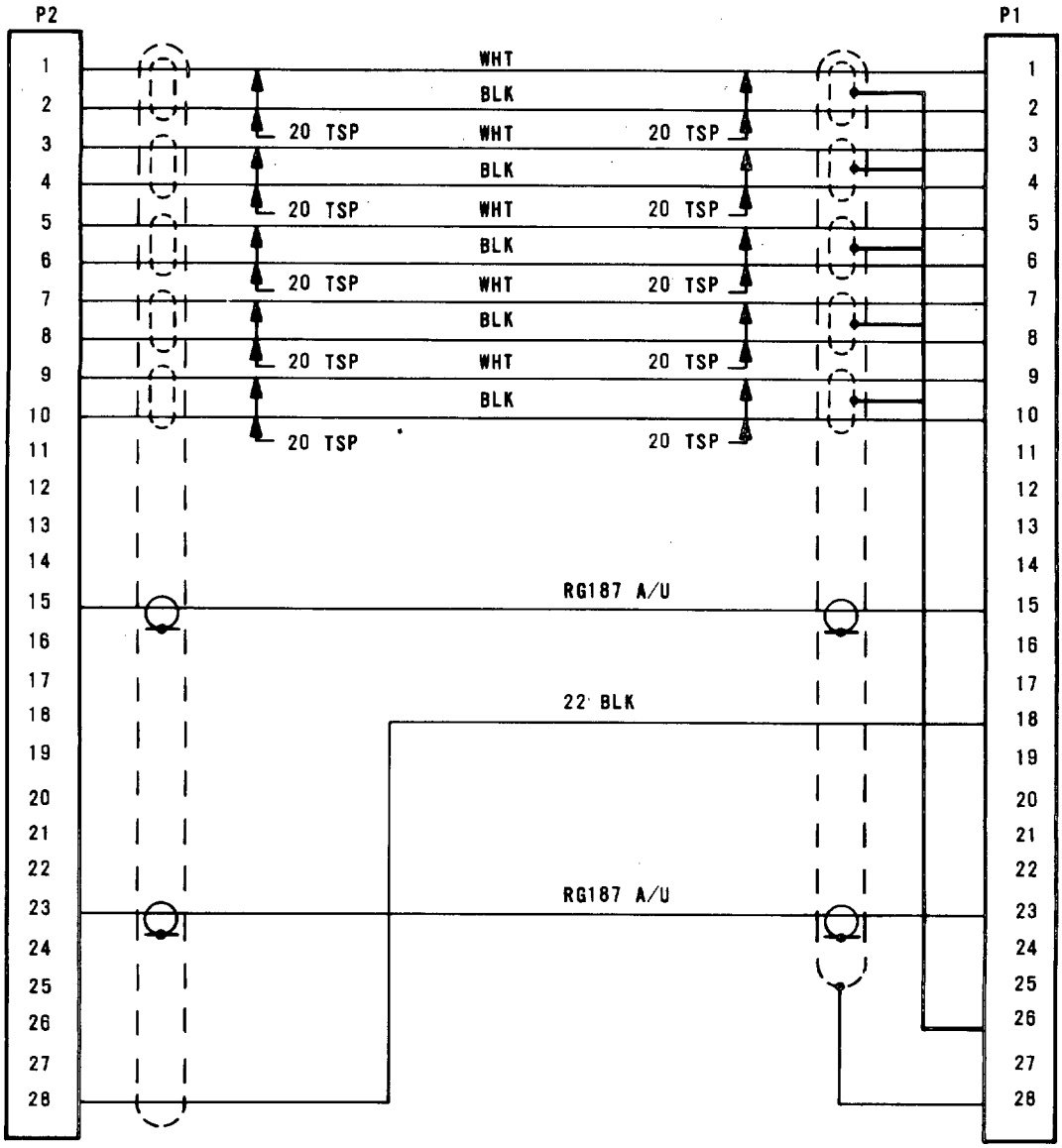


FIG. 3 INTERFACE REQUIREMENTS, CABLE ASSY W4

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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.1.6(10)

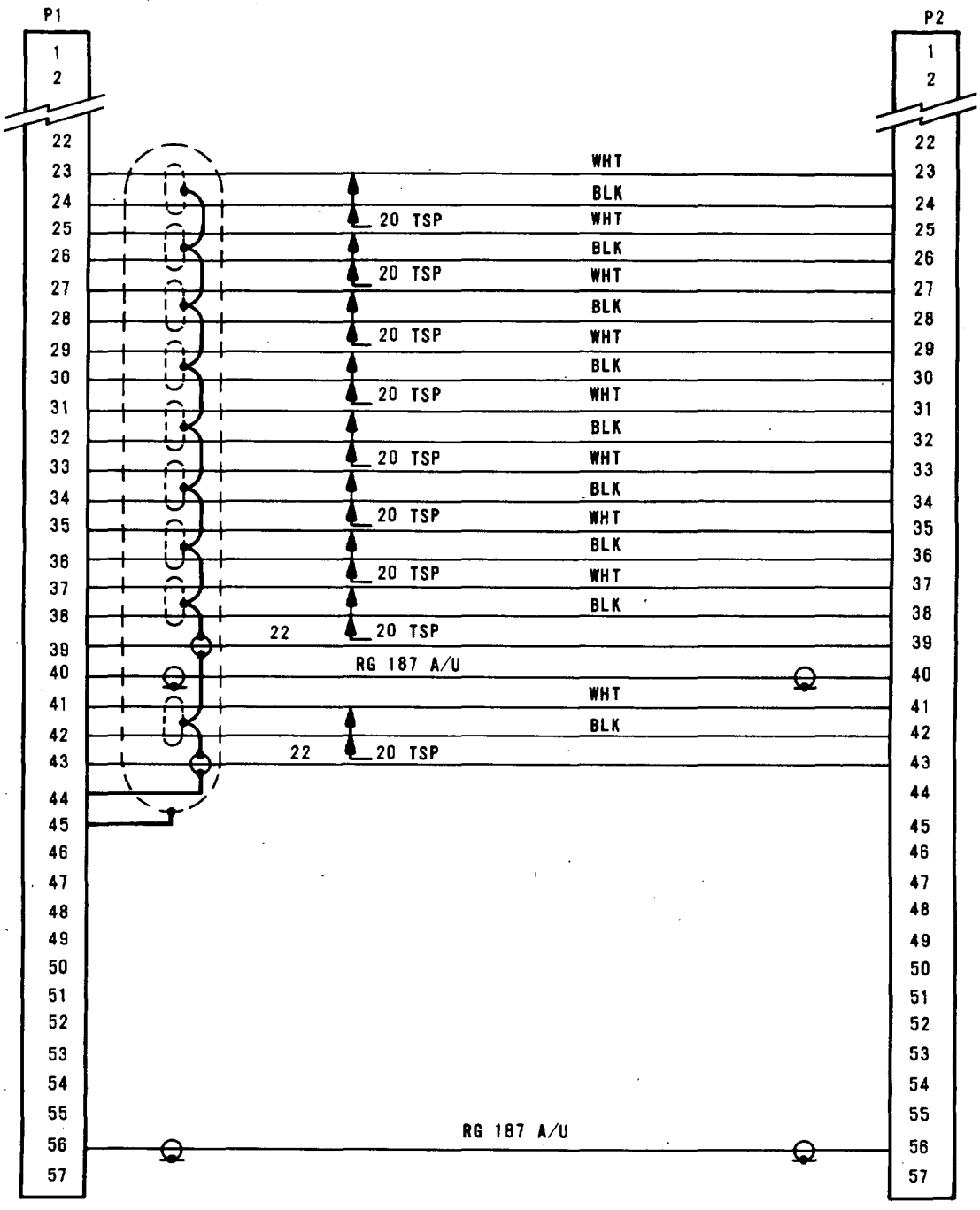
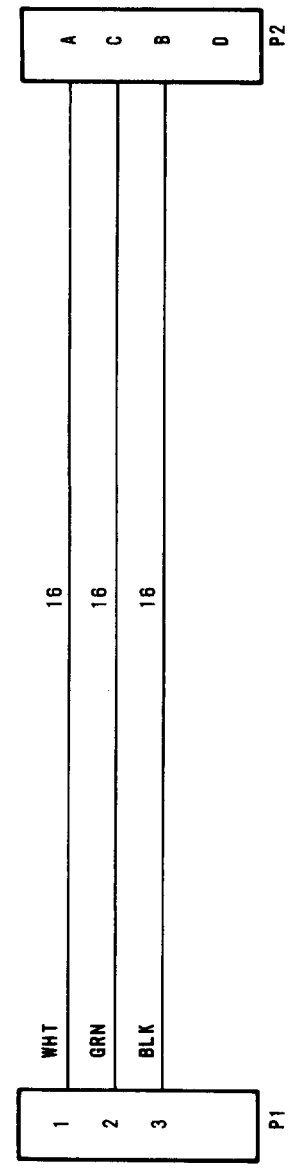


FIG. 4 INTERFACE REQUIREMENTS, CABLE ASSY W5

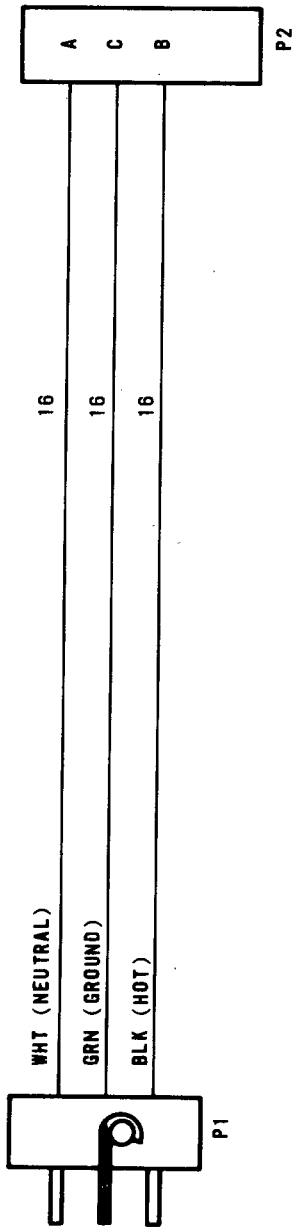
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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.1.6(10)

FIG. 5 INTERFACE REQUIREMENTS, CABLE ASSY W6

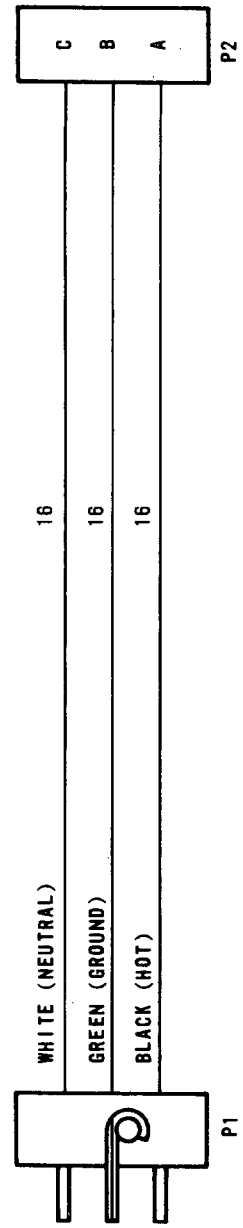
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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.1.6(10)

FIG. 6 INTERFACE REQUIREMENT, CABLE ASSY W7

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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.1.6(10)

FIG. 7 INTERFACE REQUIREMENTS, CABLE ASSY W8

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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.4.4

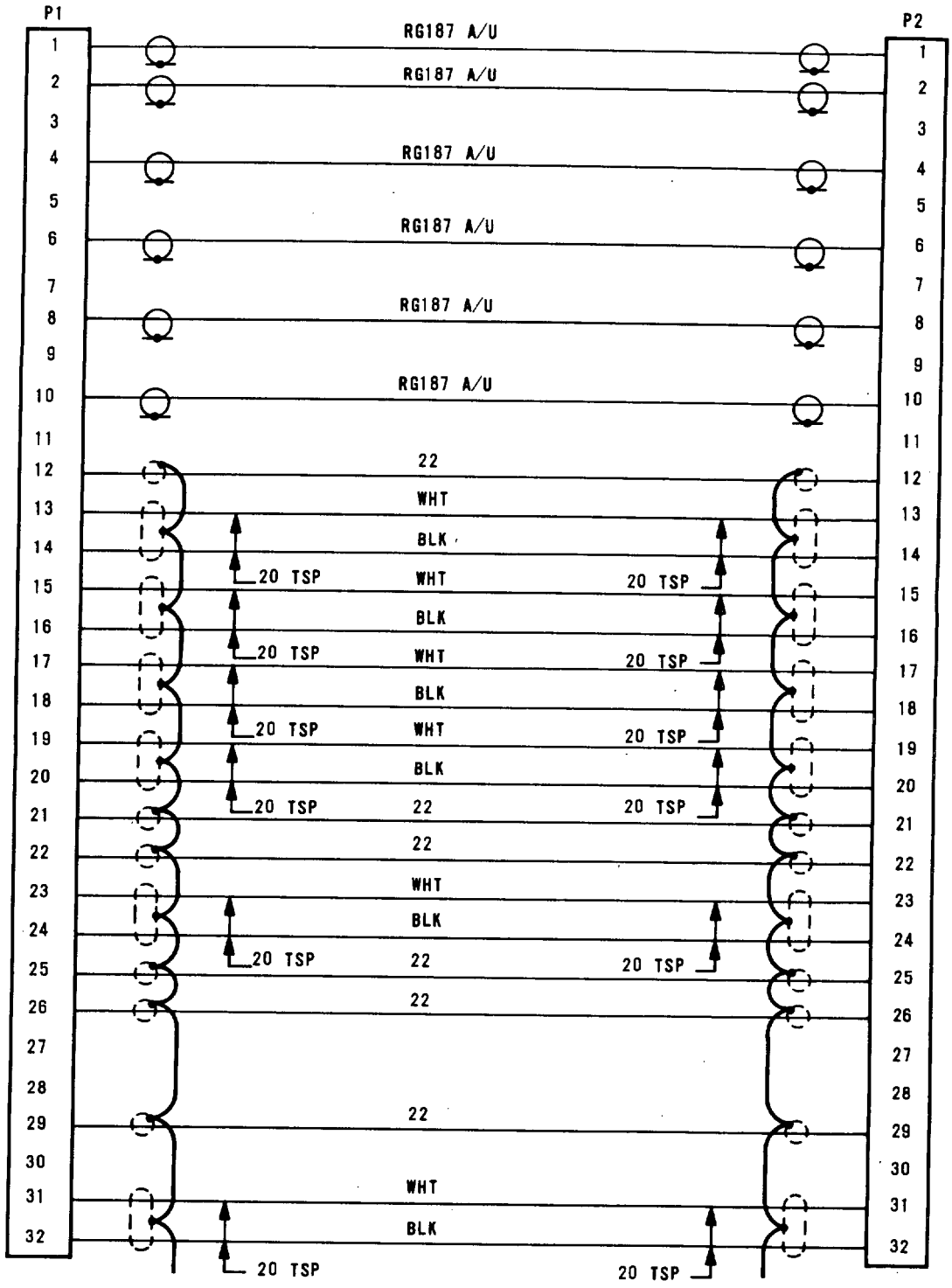


FIG. 8 INTERFACE REQUIREMENTS, CABLE ASSY A2W1

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NOTE: CONNECTORS ARE IDENTIFIED IN SECTION 3.5.4.4

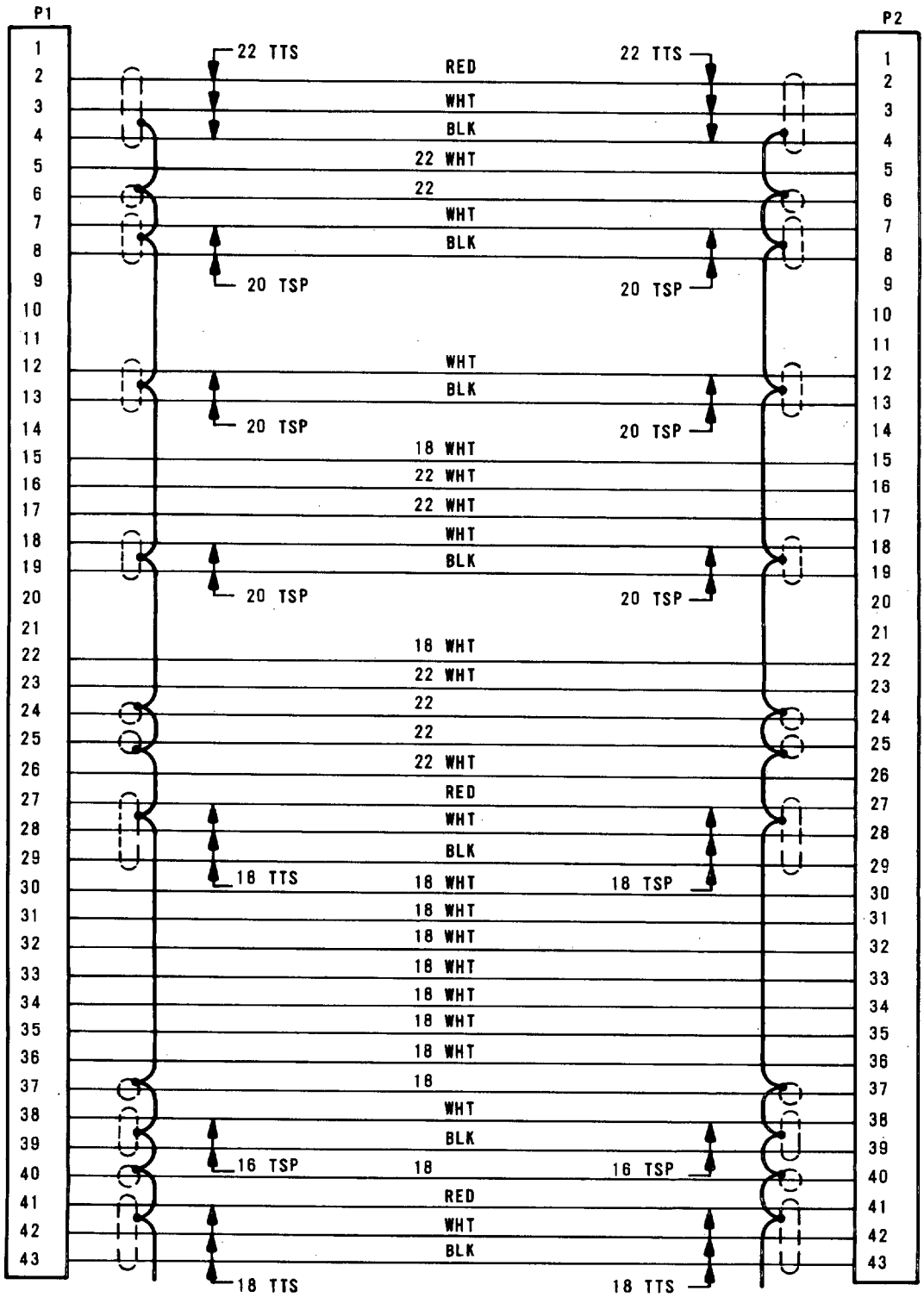


FIG. 9 INTERFACE REQUIREMENTS, CABLE ASSY A2W2

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subject to the following types of tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - When specified in the contract or order, preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests as required to demonstrate compliance with this specification.

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4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Fabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection, and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - When so specified in the contract or order, one of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

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4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted, when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitiative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Special Tests (When specified in the contract or order.)

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance withthe requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

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(1) Examination of Product

(2) Operational Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated in all modes long enough to permit the equipment temperature to stabilize and to check sufficient characteristics in accordance with Section 3 of this specification, and record adequate data to assure satisfactory equipment operation.

4.4.2 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.2.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity and the contractor.)

4.4.2.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the production tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.3 Equipment Failure - Should a failure occur during the special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

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(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - When specified in the contract or order the contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts, 60 Hz ± 5%

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the Contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

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6. NOTES

6.1 Intended Use - The equipment cabinet described in this specification is intended to support the AN/UPQ-5 Mission Recorder Display Sets and AN/USH-17(V) Mission Recorder Sets currently deployed world wide in both shipboard and shore installations. The cabinet provides mounting facilities, power and signal interface, and cooling air in a single enclosure for four classes of equipment used with the AN/UPQ-5 and AN/USH-17(V) system. First, operational installation facilities are provided for the IP-1314/UPQ-5 Video Indicator which was an addition to the AN/UPQ-5 and for which no rack space was available in the basic equipment. Second, the cabinet provides non-operating storage space for the SM-697/UPQ-5 Radar Signal Simulator and DA-662/UPQ-5 Electrical Dummy Load, two items of Peculiar Ground Support Equipment (PGSE) for the AN/UPQ-5. Third, this cabinet provides on-site storage (non-operating) for selected Shop Replaceable Assemblies (SRA's) of the AN/UPQ-5 and AN/USH-17(V) system. Fourth, the cabinet provides operational mounting facilities (including power and signal interfaces for additional equipments planned for future expansion of the mission capability of the AN/UPQ-5 and AN/USH-17(V) system. The equipments accommodated by the cabinet to support this future expansion are: 1) A Time Base Corrector to enhance the display of Electro-Optical Sensor (EOS) imagery, 2) A Video Disc Recorder which will provide a stop-action/slow motion replay capability for EOS imagery, and 3) A mission tape duplicating system comprised of a uniquely configured AN/USH-17(V) Mission Recorder, a Duplicator Interface unit, and a Static Frequency Converter.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents must specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.2.1 Procurement Options - The following is a tabulation of procurement options appearing in this specification. Purchasers should review these options for applicability. These options are invoked only when so specified in the contract or order.

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Item	Applicable Paragraph
1. Reliability Program	3.3.2
2. Mean Time Between Failure (MTBF)	3.3.2.3
3. Preproduction (First Article) Test	4.2
4. Initial Production Test	4.3
5. Special Tests	4.4
6. Life Test	4.5

6.2.2 Contract Data Requirements - Date specified in this document are not deliverable unless specified in the Contract Data Requirements List (CDRL), DD Form 1423, on the contract. The following is a tabulation of data requirements contained herein.

Item	Applicable Paragraph
1. Non-Standard Parts Data	3.2.1
2. Reliability Data	3.3.2
3. Control Panel Approval Data	3.3.4
4. Safety Data	3.3.9
5. Preproduction (First Article) Test Data	4.2.1
6. Accessory Data	4.3.2
7. Acceptance Test Reports	4.4
8. Special Test, Procedures	4.4.2.2
9. Special Test, Corrective Actions Data	4.4.2.2(3)
10. Performance Check (Life Test)	4.5.3
11. Test Data (Life Test)	4.5.4
12. Failure Reports	4.5.4.1
13. Test Procedures	4.6

6.2.2.1 Other Data Requirements - Other contract data requirements deemed appropriate by the procuring activity shall be specified in the Contract Data Requirements List (CDRL), DD Form 1423, on the contract or order.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

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(1) Contract - The contract shall have precedence over any specification. Data requirements will be identified in the contract rather than by specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified herein, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue.

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This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

(1) Mission Recorder Display Set, AN/UPQ-5

(a) Video Indicator, IP-1314/UPQ-5

(2) Radar Signal Simulator, SM-697/UPQ-5

(3) Electrical Dummy Load, DA-662/UPQ-5

(4) Mission Recorder Set, AN/USH-17(V)

(5) Static Frequency Converter, CV-2933A/UPQ-5

(6) Duplicator Interface, (Not Assigned)

(7) Time Base Corrector, (Not Assigned)

(8) Video Disc Recorder, (Not Assigned)

(9) Spare Shop Replaceable Assemblies (Listed Table I)

6.8 Metric Conversions - Table III presents metric conversion data for use with this specification.

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TABLE III METRIC CONVERSION CHART

FRACTION OF AN INCH	DECIMALS OF AN INCH	MILLIMETERS	FRACTIONS OF AN INCH	DECIMALS OF AN INCH	MILLIMETERS
1/64	0.0156	0.397	33/64	0.5156	13.097
1/32	0.0313	0.794	17/32	0.5313	13.494
3/64	0.0469	1.191	35/64	0.5469	13.891
1/16	0.0625	1.588	9/16	0.5625	14.288
5/64	0.0781	1.984	37/64	0.5781	14.684
3/32	0.0938	2.381	19/32	0.5938	15.081
7/64	0.1094	2.778	39/64	0.6094	15.478
1/8	0.1250	3.175	5/8	0.6250	15.875
9/64	0.1406	3.572	41/64	0.6406	16.272
5/32	0.1563	3.969	21/32	0.6563	16.669
11/64	0.1719	4.366	43/64	0.6719	17.066
3/16	0.1875	4.763	11/16	0.6875	17.463
13/64	0.2031	5.159	45/64	0.7031	17.859
7/32	0.2188	5.556	23/32	0.7188	18.256
15/64	0.2344	5.953	47/64	0.7344	18.653
1/4	0.2500	6.350	3/4	0.7500	19.050
17/64	0.2656	6.747	49/64	0.7656	19.447
9/32	0.2813	7.144	25/32	0.7813	19.844
19/64	0.2969	7.541	51/64	0.7969	20.241
5/16	0.3125	7.938	13/16	0.8125	20.638
21/64	0.3281	8.334	53/64	0.8281	21.034
11/32	0.3438	8.731	27/32	0.8438	21.431
23/64	0.3594	9.128	55/64	0.3594	21.828
3/8	0.3750	9.525	7/8	0.8750	22.225
25/64	0.3906	9.922	57/64	0.8906	22.622
13/32	0.4063	10.319	29/32	0.9063	23.019
27/64	0.4219	10.716	59/64	0.9219	23.416
7/16	0.4375	11.113	15/16	0.9375	23.813
29/64	0.4531	11.509	61/64	0.9531	24.209
15/32	0.4688	11.906	31/32	0.9688	24.606
31/64	0.4844	12.303	63/64	0.9844	25.003
1/2	0.5000	12.700	--	1.0000	25.400

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TABLE III METRIC CONVERSION CHART (CONT'D)

UNITS	CONVERSION FACTOR		
Millimeters	X	00.03937	= Inches
Millimeters	=	25.400	X Inches
Meters	X	3.2809	= Feet
Meters	=	00.3048	X Feet
Kilometers	X	00.621377	= Miles
Kilometers	=	1.6093	X Miles
Square centimeters	X	00.15500	= Square Inches
Square centimeters	=	6.4515	X Square Inches
Square meters	X	10.76410	= Square Feet
Square meters	=	00.09290	X Square Feet
Square kilometers	X	247.1098	= Acres
Square kilometers	=	00.00405	X Acres
Hectares	X	2.471	= Acres
Hectares	=	00.4047	X Acres
Cubic centimeters	X	.061025	= Cubic inches
Cubic centimeters	=	16.3866	X Cubic inches
Cubic meters	X	35.3156	= Cubic feet
Cubic meters	=	00.02832	X Cubic feet
Cubic meters	X	1.308	= Cubic yards
Cubic meters	=	00.765	X Cubic yards
Liters	X	61.023	= Cubic inches
Liters	=	00.01639	X Cubic inches
Liters	X	00.26418	= U.S. gallons
Liters	=	3.7854	X U.S. gallons
Grams	X	15.4324	= Grains
Grams	=	00.0648	X Grains
Grams	X	00.03527	= Ounces, avoirdupois
Grams	=	28.3495	X Ounces, avoirdupois
Kilograms	X	2.2046	= Pounds
Kilograms	=	00.4536	X Pounds
Kilograms per square centimeter	X	14.2231	= Pounds per square inch
Kilograms per square centimeter	=	00.0703	X Pounds per square inch
Kilograms per cubic meter	X	00.06243	= Pounds per cubic foot
Kilograms per cubic meter	=	16.01890	X Pounds per cubic foot
Metric tons (1,000 kilograms)	X	1.1023	= Tons (2,000 pounds)
Metric tons	=	00.9072	X Tons (2,000 pounds)
Kilowatts	X	1.3405	= Horse-power
Kilowatts	=	00.746	X Horse-power
Calories	X	3.9683	= B.T. units
Calories	=	00.2520	X B.T. units

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6.9 This specification is under the cognizance of AIR-55221B.

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